MILITARY SPECIFICATION

CABLES, SPECIAL PURPOSE, ELECTRICAL, SHIELDED

(SHIPBOARD USE)

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope. - This specification describes the requirements for the procurement of one type of electric cable suitable for high temperature, ultra high frequency use. The cable consists of a twisted pair covered by shielding braid, with an insulating jacket over the braid. The voltage rating of the cable is 600 volts root mean square (r.m.s.), and the temperature range is minus 55 to 200 degrees centigrade (C.).

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, or such portion thereof as may be designated herein, of the issue in effect on the date of invitation of bids, form a part of this specification.

SPECIFICATIONS

FEDERAL

J-C-98	Cables and Wires, Insulated; Methods of Sampling and Testing
QQ-W-343	Wire, Electrical and Non-electrical, Copper (Uninsulated)

MILITARY

MIL-C-915	Cable, Cord and Wire, Electrical (Shipboard Use)

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MIL-T-5438	Tester; Abrasion, Electrical Cable
MIL-C-12000	Cable, Cord, and Wire, Electric, Packaging and Packing of
MIL-W-16878	Wire, Electrical, Insulated, High Temperature
MIL-W-21927	Weapons, Handling and Preparation for Delivery of; General Requirements for

STANDARDS

MILITARY

MIL-STD-104	Limits for Electrical Insulation Color
MIL-STD-106	Mathematical Symbols

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN IRON AND STEEL INSTITUTE PUBLICATIONS

AISI 302 and 304	Steel Specifications - Steel Products Manual for Stainless and Heat Resisting Materials

(Application for copies should be addressed to the American Iron and Steel Institute, 150 East 42nd Street, New York 17, N. Y.)

AMERICAN INSTITUTE FOR TESTING MATERIAL PUBLICATION

ASTM B 298	Silver - Coated Soft or Annealed Copper Wire

(Application for copies should be addressed to the American Institute for Testing Materials, 1916 Race Street, Philadelphia 3, Pennsylvania.)

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AMERICAN TRUCKING ASSOCIATION, INC., PUBLICATION

Motor Freight Classification Rules

(Application for copies should be addressed to the Issuing Officer, American Trucking Association, Inc., 1424 16th Street, N.W., Washington 6, D. C.)

CONSOLIDATED CLASSIFICATION COMMITTEE PUBLICATION

Consolidated Freight Classification Rules

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 General requirements.

3.1.1 Pre-production samples. - Unless otherwise specified in the contract or order, pre-production samples of the control motor shall be manufactured using the methods and procedures proposed for the production lot. These samples will be tested as specified in section 4 herein and are for the purpose of determining that, prior to starting production, the contractor's production methods are capable of yielding items that comply with the technical requirements of the contract. Provision of samples shall be in accordance with 4.1.1.

3.1.2 Materials.

3.1.2.1 Quality. - All material used in the construction of completed cable shall be of the quality and form best suited for the purpose intended.

3.1.2.2 Cable components.

3.1.2.2.1 Conductors. - The copper strands of the conductors shall be soft or drawn-and-annealed copper strands, coated with commercially pure silver for a coating thickness not less than 40 microinches. The steel strands of the conductor (when steel strands are used) shall be of steel meeting the requirements of AISI 302 or AISI 304.

3.1.2.2.2 Conductor insulation. - The conductor insulation shall be of polytetrafluoroethylene.

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3.1.2.2.3 Shielding braid. - The strands of the shielding braid shall be of soft-drawn copper wire, coated with commercially pure silver for a coating thickness of not less than 40 microinches.

3.1.2.2.4 Jacket. - The cable jacket shall be of white polytetrafluoroethylene.

3.1.3 Gages. - The contractor shall provide himself with whatever gages are necessary and adequate to ensure that the material to which this specification applies will meet the requirements shown on the applicable drawings. If so stipulated in the contract, order, or requisition, the Government will furnish drawings of pertinent Navy Final Inspection Gages for guidance and design of the contractor's inspection gages. However, such procedure shall not relieve the contractor of his responsibility in the design and manufacture of such gages as may be required for the satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of the cable by Navy Final Inspection Gages.

3.1.4 Requests for deviation. - Each request that a deviation be authorized shall have been submitted to the Procuring Activity, and the deviation shall have been authorized by the Bureau of Naval Weapons or by a Naval Activity delegated such authority, prior to the production of the affected item, or prior to the performance of a particular operation on the item. The deviation shall apply only to this specific item or operation. The request shall define the suggested deviation in sufficient detail to permit the Procuring Activity to evaluate its effect on performance and reliability, and the reasons for the suggested deviation shall be stated. Each request that a deviation be authorized, and the authorization, shall state specifically the extent of the departure, the identifying number of the item, the contractor (or bidder) and the contract number. A deviation is essentially temporary in character, and usually represents a departure limited to a specific period, to a specific number of items, or to the contract to which it applies. It may require an equitable reduction in contract price if a decrease in the cost of performance results therefrom.

3.1.5 Conflicting requirements. - Conflicting requirements arising between this specification and any specification, publication, or drawing listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation, clarification, and resolution (or correction).

3.2 Definitions. - The definitions of terms in Specification J-C-98 and Specification MIL-C-915 shall be applicable to this specification.

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3.3 Performance requirements and product characteristics. - The cable shall meet the following performance requirements and product characteristics:

3.3.1 Conductor resistance. - When measured in accordance with 4.4.4.1, the resistance of each conductor of the cable shall meet the requirements of table I. See 4.4.4.1.

3.3.2 Dielectric strength and insulation resistance.

3.3.2.1 Dielectric strength. - When tested in accordance with 4.4.4.2.1, the cable shall withstand an alternating-current (a-c) potential of 2000 volts root-mean-square (r.m.s.) having approximately sinusoidal waveform and frequency between 25 and 100 cycles per second (c.p.s.) for 60 seconds without arcing, damage, or breakdown. See 4.4.4.2.1.

3.3.2.2 Insulation resistance. - When measured in accordance with 4.4.4.2.2, the insulation resistance in megohms shall not be less than 1,000,000 divided by the length of the cable in feet (i.e. the insulation resistance expressed in megohms per 1000 feet shall not be less than 1000). See 4.4.4.2.2.

3.3.3 Visual and dimensional requirements.

3.3.3.1 Conductors.

3.3.3.1.1 Number of strands. - When checked in accordance with 4.5.1.1.1, the number of copper strands and steel strands in each conductor shall be as specified in table I. See 4.5.1.1.1.

3.3.3.1.2 Position of steel strands. - When checked in accordance with 4.5.1.1.2, the position of the steel strands in each conductor shall be as follows:

(a) The center strand shall be the steel strand in cable sizes AWG 26, 28, 30, and 32,

(b) The center strand and two diametrically opposite strands next to it shall be the steel strands for cable sizes AWG 20, 22, and 24,

(c) There shall be no steel strands in cable sizes AWG 16 and 18.

See 4.5.1.1.2.

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Table I. Cable Dimensions

Cable Size (AWG)	Conductors									Shielding Braid				Finished Cable		
	No. of Copper Strands	No. of Steel Strands	Diameter of Strands (Nominal)	Cross Sectional Strands (Nominal)	Diameter Over Strands (Nominal)		Diameter Over Insulation (Max.)	Average Insulation Thickness (Min.)	Diameter Over Core (Nominal)	Diameter of Strands (Nominal)	No. of Carriers	Ends per Carrier	Picks per Inch	Diameter Over Jacket		Conductor Resist. per 1000 Feet at 20°C (Max.)
				Circular Mils	Outer Layer	Inner Layer								Min.	Max.	
			Inch		Inch	Inch	Inch	Inch	Inch	Inch				Inch	Inch	Ohms
32	6	1	.0031	67.3	.009	-	.033	.008	.058	.0040	16	4	25	.091	.101	205
30	6	1	.0040	112.0	.012	-	.036	.008	.064	.0040	16	4	25	.097	.107	123
28	6	1	.0050	175	.015	-	.039	.008	.070	.0040	16	4	23	.103	.113	78.7
26	6	1	.0063	278	.019	-	.043	.008	.078	.0050	16	4	19	.124	.126	49.5
24	16	3	.0050	475	.025	.015	.049	.008	.090	.0050	16	4	18	.126	.138	27.0
22	16	3	.0063	754	.032	.019	.056	.008	.104	.0050	16	5	17	.139	.153	18.3
20	16	3	.0080	1216	.040	.024	.064	.008	.120	.0050	16	5	16	.154	.170	11.3
18	19	0	.0100	1900	.050	.030	.074	.008	.140	.0050	16	6	14	.173	.191	7.26
16	19	0	.0113	2426	.057	.034	.083	.009	.158	.0050	16	7	12	.190	.210	5.68

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3.3.3.1.3 Diameter of strands. - When measured in accordance with 4.5.1.1.3, the diameter of the strands of each conductor shall be in accordance with table I with the tolerances of table II applicable. See 4.5.1.1.3.

Table II. Strand Diameter Tolerances

Nominal Diameter	Tolerances
Under 10 mils	+ 0.3 mil - 0.1 mil
10 mils and over	+ 3 percent - 1 percent

3.3.3.1.4 Cross-sectional area of stranding. - When determined in accordance with 4.5.1.1.4, the cross sectional area of the stranding of each conductor shall be not less than 98 percent of the cross-section area specified in table I for the specified AWG size of the cable being checked. See 4.5.1.1.4.

3.3.3.1.5 Direction of lay of strands. - When checked in accordance with 4.5.1.1.5, the direction of lay shall be left-hand for the outer layer of strands of each conductor. See 4.5.1.1.5.

3.3.3.1.6 Length of lay of strands. - When measured in accordance with 4.5.1.1.6, the length of lay of each layer of strands of each conductor shall not be less than 8 nor more than 16 times the nominal diameter of that layer as specified in table I. See 4.5.1.1.6.

3.3.3.1.7 Construction of stranding. - When checked in accordance with 4.5.1.1.7, each conductor shall be of concentric or unilay construction. See 4.5.1.1.7.

3.3.3.1.8 Color of insulation. - When checked in accordance with 4.5.1.1.8, the color of the insulation of one conductor shall be black, the color of the insulation of the other conductor shall be white, and the color of the insulation of each conductor shall be within the applicable color limits given in Standard MIL-STD-104. See 4.5.1.1.8.

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3.3.3.1.9 Application of insulation. - When the insulation of each conductor is checked in accordance with 4.5.1.1.9, it shall consist of a tight fitting tube of uniform thickness of polytetrafluoroethylene which is extruded or taped on. If the insulation is taped on, the convolutions shall be fused to form a smooth homogenious tube, but a slight spiral ridge shall not be considered cause for rejection. See 4.5.1.1.9.

3.3.3.1.10 Diameter over insulation. - When measured in accordance with 4.5.1.1.10, the diameter of each conductor over the insulation shall meet the requirements of table I. See 4.5.1.1.10.

3.3.3.1.11 Insulation thickness.

3.3.3.1.11.1 Average thickness. - When measured in accordance with 4.5.1.1.11.1, the average thickness of the insulation of each conductor shall meet the requirements of table I. See 4.5.1.1.11.1.

3.3.3.1.11.2 Minimum thickness. - When measured in accordance with 4.5.1.1.11.2, the minimum thickness of the insulation of each conductor shall not be less than 70 percent of the maximum thickness, both measurements being made at the same cross section. See 4.5.1.1.11.2.

3.3.3.1.12 Length of lay of conductors. - When measured in accordance with 4.5.1.1.12, the length of lay of the twisted conductors shall not be less than 8 nor more than 16 times the nominal core diameter (i.e. the diameter over the twisted conductors) given in table I. See 4.5.1.1.12.

3.3.3.1.13 Fillers and tape. - When checked in accordance with 4.5.1.1.13, no fillers shall be used in the interstices, and no cable tape shall be used. See 4.5.1.1.13.

3.3.3.2 Shielding braid.

3.3.3.2.1 Embedding in conductor insulation. - When checked in accordance with 4.5.1.2.1, the extent of any embedding of the shielding braid in the insulation of the twisted conductors shall be such that the embedding does not noticeably interfere with pushing back the shielding braid over the conductors. See 4.5.1.2.1.

3.3.3.2.2 Opening. - When checked in accordance with 4.5.1.2.2, the diameter the shielding braid opens to when it is pushed back over the twisted conductors shall be approximately 0.05 inch but not less than 0.04 inch greater than the diameter over the twisted conductors. See 4.5.1.2.2.

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3.3.3.2.3 Diameter of strands. - When measured in accordance with 4.5.1.2.3, the diameter of the shielding braid strands shall be in accordance with table I, with the tolerances of table II applicable. See 4.5.1.2.3.

3.3.3.2.4 Coverage. - When checked in accordance with 4.5.1.2.4, the coverage of the shielding braid shall not be less than 80 percent. See 4.5.1.2.4.

3.3.3.2.5 Embedding in jacket. - When checked in accordance with 4.5.1.2.5, the extent of any embedding of the shielding braid in the jacket shall be such that when a section of jacket is removed from the cable and the finger nail is moved along the inner surface of the section of jacket, lengthwise, and perpendicular to both the surface of the jacket and the direction of motion, the fingernail shall not stick in any indentation caused by embedding. See 4.5.1.2.5.

3.3.3.3 Jacket.

3.3.3.3.1 Color. - When checked in accordance with 4.5.1.3.1, the color of the jacket shall be white and shall be within the applicable color limits given in Standard MIL-STD-104. See 4.5.1.3.1.

3.3.3.3.2 Application. - When checked in accordance with 4.5.1.3.2, the jacket shall consist of a tight-fitting tube of uniform thickness of white polytetrafluoroethylene which is extruded or taped on. If the jacket is taped on, it shall consist of two layers of tape applied in opposite directions with the two layers and the convolutions fused to form a smooth homogenious tube. However, a slight spiral ridge on either surface of the jacket shall not be considered cause for rejection. See 4.5.1.3.2.

3.3.3.3.3 Diameter over jacket. - When measured in accordance with 4.5.1.3.3, the diameter of the finished cable over the jacket shall meet the requirements of table I. See 4.5.1.3.3.

3.3.3.3.4 Average thickness. - When measured in accordance with 4.5.1.3.4, the average thickness of the jacket shall be 0.008 inch minimum. See 4.5.1.3.4.

3.3.4 Strand coating and tensile requirements.

3.3.4.1 Continuity of coating (copper strands). - When checked in accordance with 4.5.2.1, the silver coating of the copper strands of the conductors and shielding braid shall be free of visible discontinuities. See 4.5.2.1.

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3.3.4.2 Thickness of coating (copper strands). - When measured in accordance with 4.5.2.2, the thickness of the silver coating of the copper strands of the conductors and shielding braid shall be 40 microinches minimum. See 4.5.2.2.

3.3.4.3 Tensile properties.

3.3.4.3.1 Copper strands. - When the silver plated copper strands of the conductors and shielding braid are tested individually in accordance with 4.5.2.3.1, the minimum tensile strength and maximum elongation (in a 10-inch gage length) shall be respectively 40,000 pounds per square inch and 10 percent. See 4.5.2.3.1.

3.3.4.3.2 Steel strands. - When the steel strands of the conductors are tested individually in accordance with 4.5.2.3.2, the minimum tensile strength and maximum elongation (in a 10-inch gage length) shall be respectively 144,000 pounds per square inch and 10 percent. See 4.5.2.3.2.

3.4 Environmental requirements. - The cable shall meet the other applicable requirements of this specification during or after tests defined or specified in 4.5.3.

3.4.1 Cold bending. - The cable shall withstand the cold bending test of 4.5.3.1 and thereafter shall show no damage and shall meet the requirements of 3.3.2.1. See 4.5.3.1.

3.4.2 Heat resistance. - The cable shall withstand the heat resistance test of 4.5.3.2, and thereafter shall show no damage and shall

(a) meet the requirements of 3.3.2.1,
(b) not have shrunk or elongated more than 1/8 inch,
(c) meet the requirements of 3.3.3.1.8 and 3.3.3.3.1.

See 4.5.3.2.

3.4.3 Soldering resistance. - The cable shall withstand the soldering resistance test of 4.5.3.3, and thereafter shall show no damage and shall not have shrunk or elongated more than 1/16 inch. See 4.5.3.3.

3.4.4 Abrasion resistance. - The cable shall withstand the abrasion resistance test of 4.5.3.4, i.e. the abrasion resistance of the conductor insulation shall be such that when 8 readings of inches of tape travel to failure are obtained in accordance with 4.5.3.4, the arithmetic average of the readings which are less than the arithmetic average of all 8 readings shall not be less than 17 inches. See 4.5.3.4.

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3.4.5 Shock-transportation, storage, and handling. - The complete pack after being subjected to the tests specified in 4.5.3.5 shall not rupture so as to spill contents or break seals on unit package.

3.5 Clean-up. - Prior to and after final assembly, all components and the cable shall be thoroughly cleaned of wire fragments and other foreign matter.

3.6 Workmanship. - The cable, including all components, shall be constructed and finished in a manner to assure compliance with the requirements of this specification. The workmanship shall be in accordance with high grade commercial practice covering this type of material.

3.6.1 The workmanship exhibited in the pre-production sample will be evaluated to determine acceptability. The approved standards of workmanship will thereby become a minimum production requirement for all units offered for acceptance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Pre-production sample. - Unless otherwise specified for each contract or order, a representative sample of 200 feet of cable of each AWG size listed in table I shall be manufactured by the methods and procedures proposed for the production lot. These samples shall be submitted to the Government inspector for pre-production tests at an activity designated by the Bureau of Naval Weapons, for purposes of determining compliance with the requirements of the contract, specifications, and drawings. Further production of the item by the contractor prior to approval of the contracting activity or the completion of tests on the pre-production samples shall be at the contractor's risk. Pre-production samples accepted will be applied as part of the quantity specified by the contract or order, but shall not be used in any equipment manufactured for the Government.

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4.1.2 Periodic production samples. - Randomly select from each production lot unit packages of cable in accordance with tables III and IV. Unless otherwise specified in the contract or purchase order, these units shall be submitted to a Bureau of Naval Weapons assigned activity to determine compliance of the samples with the requirements of the contract, specifications, and drawings. The unit packages of cable shall be considered satisfactory if they meet all specifications and pertinent drawing requirements and withstand the tests specified in 4.4 and 4.5. Periodic production samples accepted will be applied as part of the quantity specified by the contract or order, but shall not be used in any equipment manufactured for the Government.

Table III

Sampling for visual and dimensional examination and for strand coating and tensile tests

Number of Spools or Reels in Inspection Lot	Number of Spools or Reels in Sample	Acceptance Number (defectives)	Rejection Number (defectives)
15 or under	5	0	1
16 to 40	7	0	1
41 to 110	10	0	1
111 to 300	15	1	2
301 and over	25	1	2

Table IV

Sampling for environmental tests

Number of Spools or Reels in Inspection Lot	Number of Spools or Reels in Sample	Acceptance Number (defectives)	Rejection Number (defectives)
40 or under	2	0	1
41 to 110	3	0	1
111 to 300	4	0	1
301 and over	5	0	1

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4.2 Classification of tests. - The inspection and testing of the cable shall be classified as follows:

4.2.1 Acceptance tests. - These tests shall be accomplished on each unit package of cable submitted for acceptance under contract. These tests are detailed in 4.4.

4.2.2 Pre-production and periodic production tests. - Pre-production and periodic production tests shall be accomplished on samples selected as specified in 4.1.1 and 4.1.2 which are representative of the production of the cable after the award of the contract, to determine that the production meets the requirements of this specification. These tests are detailed in 4.5.

4.2.2.1 Action in case of failure.

4.2.2.1.1 Pre-production tests. - Acceptance shall be based on no failure to meet the tests of 4.5, and no other defects. Failure of the sample to comply with these requirements will result in rejection of the sample or the cessation of production, as determined by the procuring activity.

4.2.2.1.2 Periodic production tests. - Acceptance shall be based on no more failures to meet the tests of 4.5 than are permitted in tables III and IV, and no other defects. Failure to comply with these requirements shall result in rejection of the lot or the cessation of production, as determined by the procuring activity.

4.2.3 Tests after delivery. - Unless otherwise specified in the contract, the Government reserves the right to conduct any or all the tests specified herein, as deemed necessary, after installation of the cable. Any defects discovered during or after these tests due to the contractor's failure to comply with drawings or specifications for this equipment shall be corrected by the contractor at no expense to the Government, provided the contractor is notified of such defects within the guarantee period of the contract.

4.3 Test equipment. - The following items of test equipment are required to perform the acceptance tests set forth in this specification.

4.3.1 Thermometer. - A thermometer calibrated in degrees centigrade, having an accuracy of plus or minus 0.5 degrees centigrade.

4.3.2 Wheatstone bridge. - A direct-current resistance bridge capable of measuring resistance from 1 to 30,000 ohms with an accuracy of plus or minus 0.25 percent.

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4.3.3 High-potential tester. - A high-potential tester capable of applying up to 2000 volts r.m.s., 60 c.p.s., and having a volt-ampere rating of 1000 minimum. When the unit is turned on or off, the output voltage shall change uniformly with time, and shall reach 100 percent or zero percent rated output in not less than 10 nor more than 60 seconds. The unit shall have a meter for reading dielectric current, and shall cut out when the dielectric current exceeds 2 milliamperes.

4.3.4 Insulation resistance tester. - An insulation tester capable of measuring resistance to 100,000 megohms, using a direct-current test voltage not less than 100 nor greater than 500 volts.

4.4 Acceptance tests.

4.4.1 The contractor shall conduct acceptance tests as specified in 4.2.1 to assure that the cable is in compliance with the requirements of 3.1, 3.3.1, 3.3.2, 3.5, and 3.6. Manufacturer's certification of materials specified in the applicable drawings shall be used in lieu of material tests.

4.4.2 Test conditions. - Unless otherwise specified, the cable shall be subjected to acceptance tests under the following conditions:

4.4.2.1 Temperature: Room ambient 65 to 95 degrees Fahrenheit (18 to 35 degrees centigrade).

4.4.2.2 Altitude: Normal ground atmospheric pressure.

4.4.2.3 Vibration: None.

4.4.2.4 Humidity: Room ambient to 95 percent relative maximum.

4.4.3 Test and inspection equipment and facilities. - The manufacturer shall furnish and maintain all necessary test equipment, facilities, and personnel for performing all acceptance tests. The test equipment shall be adequate in quantity and, unless definite requirements are specified, shall be of sufficient accuracy and quality to permit performance of the required acceptance tests.

4.4.4 Test procedure. - The detailed test procedure listed in the following paragraphs shall be followed.

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4.4.4.1 Conductor resistance. - Measurement of conductor resistance shall not be made until the temperature of the spool of cable has stabilized at room ambient temperature. The resistance of the entire length of cable on the spool (each conductor separately) shall then be measured using the Wheatstone bridge of 4.3.2, and the resistance of the cable in ohms per 1000 feet at 20 degrees centigrade shall be computed as follows to determine compliance with the requirements of 3.3.1.

$$R = \frac{1000\ R_T}{L\ [1 + 0.00393(T-20)]}$$

where

R = ohms per 1000 feet at 20 degrees centigrade,

R_T = resistance of cable conductor at temperature T,

T = room temperature (degrees centigrade),

L = length of cable in feet as marked on the spool by the cable manufacturer

Mathematical symbols shall be interpreted in accordance with Standard MIL-STD-106.

4.4.4.2 Dielectric strength and insulation resistance.

4.4.4.2.1 Dielectric strength. - A cable specimen 20 feet plus or minus 0.5 foot in length shall be used for this test. A potential of 2000 volts r.m.s., 60 c.p.s. shall be applied between conductors - first for one conductor connected to the shielding braid, and then for the other conductor connected to the shielding braid, using the high-potential tester of 4.3.3. The high potential tester shall be turned off 60 seconds after the high-potential voltage reaches 2000 volts r.m.s.

To determine compliance with 3.3.2.1, the cable specimen and the current meter of the high-potential tester shall be observed for any indication of arcing during application of high-potential voltage (any abrupt change in current reading indicates arcing), and the cable specimen shall be inspected visually for evidence of arcing after application of the high-potential voltage.

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4.4.4.2.2 Insulation resistance. - Immediately following the dielectric strength test of 4.4.4.2.1, the insulation resistance shall be measured between conductors - first for one conductor connected to the shielding braid, and then for the other conductor connected to the shielding braid, using the insulation resistance tester of 4.3.4, to determine compliance with the requirements of 3.3.2.2.

4.4.5 Preparation for delivery. - Preparation for delivery of the cables shall conform to section 5 of this specification.

4.5 Pre-production and periodic production sample tests.

4.5.1 Visual and dimensional requirements.

4.5.1.1 Conductors.

4.5.1.1.1 Number of strands. - The number of copper strands and steel strands in each conductor shall be counted to determine compliance with the requirements of 3.3.3.1.1.

4.5.1.1.2 Position of steel strands. - The position of the steel strands in each conductor shall be observed to determine compliance with the requirements of 3.3.3.1.2.

4.5.1.1.3 Diameter of strands. - The diameter of the strands of each conductor shall be measured using the method specified for the measurement of solid wire and strand diameters in Specification QQ-W-343 to determine compliance with the requirements of 3.3.3.1.3.

4.5.1.1.4 Cross-sectional area of stranding. - The cross-sectional area of the stranding of each conductor shall be computed using the method specified for computing cross-sectional area of stranded wire in Specification QQ-W-343 to determine compliance with the requirements of 3.3.3.1.4.

4.5.1.1.5 Direction of lay of strands. - The direction of lay of the outer layer of strands of each conductor shall be observed to determine compliance with the requirements of 3.3.3.1.5.

4.5.1.1.6 Length of lay of strands. - The length of lay of each layer of strands of each conductor shall be measured using a steel scale graduated to 1/32 inch or finer to determine compliance with the requirements of 3.3.3.1.6.

4.5.1.1.7 Construction of stranding. - The construction of the stranding shall be checked visually to determine compliance with the requirements of 3.3.3.1.7.

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4.5.1.1.8 Color of insulation. - The color of the insulation of each conductor shall be checked visually against the applicable color limit given in Standard MIL-STD-104 to determine compliance with the requirements of 3.3.3.1.8.

4.5.1.1.9 Application of insulation. - The insulation of each conductor shall be checked visually to determine compliance with the requirements of 3.3.3.1.9.

4.5.1.1.10 Diameter over insulation. - The diameter of each conductor over the insulation shall be measured using the method specified for measurement of solid wire and strand diameters in Specification QQ-W-343 to determine compliance with the requirements of 3.3.1.10.

4.5.1.1.11 Insulation thickness.

4.5.1.1.11.1 Average thickness. - The average thickness of the insulation of each conductor shall be measured using method 1011, 1013, or 1018 of Specification J-C-98 to determine compliance with the requirements of 3.3.3.1.11.1.

4.5.1.1.11.2 Minimum thickness. - The minimum and maximum thickness of the insulation of each conductor shall be measured at the same cross section, and the ratio of minimum thickness to maximum thickness shall be computed to determine compliance with the requirements of 3.3.3.1.11.2. The minimum thickness of insulation shall be measured in accordance with method 1011, 1013, or 1018 of Specification J-C-98.

If method 1011 or 1013 of Specification J-C-98 is used, the diameter over the insulated conductor at the point where the insulation is considered thinnest shall be measured before removing the insulation on the side opposite this point so that the maximum insulation thickness at the cross section can also be computed, i.e. the maximum insulation thickness is the diameter over the insulated conductor at the point considered thinnest minus the same measurement made with the insulation removed from the side of the conductor opposite this point.

If method 1018 is used, the maximum insulation thickness shall be determined using the same procedure specified for determining the minimum insulation thickness.

4.5.1.1.12 Length of lay of conductors. - The length of lay of the twisted conductors shall be measured using a steel scale graduated to 1/32 inch or finer to determine compliance with the requirements of 3.3.3.1.12.

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4.5.1.1.13 Fillers and tape. - The twisted conductors shall be inspected visually to determine compliance with the requirements of 3.3.3.1.13.

4.5.1.2 Shielding braid.

4.5.1.2.1 Embedding in conductor insulation. - With the jacket removed from at least 6 inches of cable, the shielding braid shall be pushed back manually to determine compliance with the requirements of 3.3.3.2.1.

4.5.1.2.2 Opening. - A right cylindrical section of shielding braid 2 inches plus or minus 1/8 inch long, shall be removed from the cable, and the diameter over the twisted conductors where the shielding braid was removed shall be measured using a dial micrometer or micrometer caliper graduated to read in mils (0.001 inch). The 2-inch section of shielding braid shall be compressed lengthwise enough to cause it to open to its maximum diameter, and it shall be determined whether the section of shielding braid thus compressed will fit over a plug gage having diameter 0.040 inch greater than the measured diameter over the twisted conductors to determine compliance with the requirements of 3.3.3.2.2.

4.5.1.2.3 Diameter of strands. - The diameter of the strands of the shielding braid shall be measured using the method specified for the measurement of solid wire and strand diameters in specification QQ-W-343 to determine compliance with the requirements of 3.3.3.2.3.

4.5.1.2.4 Coverage. - The number of carriers, ends per carrier, and picks per inch shall be ascertained using method 8011 and 8021 of Specification J-C-98. If all of the following conditions are met,

(a) the number of carriers, ends per carrier, and picks per inch are in accordance with table I,
(b) the requirements of 3.3.3.1.10 are satisfied,
(c) the requirements of 3.3.3.2.3 are satisfied,

then the coverage shall be considered to be at least 80 percent. If not all of the above conditions are met, then the coverage shall be computed using the formula for percent coverage given in Specification MIL-W-16878 to determine compliance with the requirements of 3.3.3.2.4.

4.5.1.2.5 Embedding in jacket. - A section of jacket several inches long shall be removed from the cable, and shall be cut open lengthwise. The finger nail or thumb nail shall then be moved along the inner surface of the section of jacket as specified in 3.3.3.2.5 to determine compliance with the requirements of 3.3.3.2.5.

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4.5.1.3 Jacket.

4.5.1.3.1 Color. - The color of the jacket shall be checked visually against the color limit for white given in Standard MIL-STD-104 to determine compliance with the requirements of 3.3.3.3.1.

4.5.1.3.2 Application. - A section of jacket several inches long, removed from the cable and cut open lengthwise, shall be examined to determine compliance with the requirements of 3.3.3.3.2.

4.5.1.3.3 Diameter over jacket. - The diameter of the finished cable over the jacket shall be measured using the method specified for measurement of solid wire and strand diameters in Specification QQ-W-343 to determine compliance with the requirements of 3.3.3.3.3.

4.5.1.3.4 Average thickness. - A short length of jacket shall be removed from the cable, the thickness of the length of jacket shall be measured at two diametrically opposite points at the same cross section using a pin gage micrometer meeting the requirements of the pin gage micrometer specified in method 1014 of Specification J-C-98, and one-half the sum of the two thickness measurements shall be computed to determine compliance with the requirements of 3.3.3.3.4.

4.5.2 Strand coating and tensile requirements.

4.5.2.1 Continuity of coating (copper strands). - The continuity of the silver coating of the copper strands of conductors and shielding braid shall be checked in accordance with ASTM B 298 to determine compliance with the requirements of 3.3.4.1.

4.5.2.2 Thickness of coating (copper strands). - The thickness of the silver coating of the copper strands of the conductors and shielding braid shall be measured in accordance with ASTM B 298 to determine compliance with the requirements of 3.3.4.2.

4.5.2.3 Tensile properties.

4.5.2.3.1 Copper strands. - The tensile strength and elongation of the silver plated copper strands of the conductors and shielding braid shall be measured in accordance with the procedure given for measuring the tensile strength and elongation of soft or drawn-and-annealed solid wire and strands in Specification QQ-W-343 (except that the specified measurements need not be made before stranding or braiding) to determine compliance with the requirements of 3.3.4.3.1.

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4.5.2.3.2 Steel strands. - The tensile strength and elongation of the steel strands of the conductors shall be measured in accordance with the procedure given for measuring the tensile strength and elongation of medium-hard-drawn and hard-drawn solid wires and strands in Specification QQ-W-343 to determine compliance with the requirements of 3.3.4.3.2.

4.5.3 Environmental requirements.

4.5.3.1 Cold bending. - A cable specimen having a length of 12 inches greater than the length required for winding on the applicable mandrel specified in table V shall be used for this test. The specimen shall be secured to the mandrel in such a manner that bending will occur at a distance of at least 6 inches from one end of the specimen. A minimum of two turns shall be made on the mandrel at room temperature. The free end of the specimen shall be secured to the load weight specified in table V. The entire coiled specimen shall be placed in a cold chamber at a temperature of minus 73 degrees centigrade (minus 100 degrees Fahrenheit) for a period of 4 hours. While in the cold chamber at this temperature, the mandrel shall be rotated at an approximate rate of one turn per 4 seconds until a total of 6 turns of wire have been wound on the mandrel. The 6 turns of wire shall then be unwound from the mandrel and rewound onto the mandrel in a direction opposite to the direction of the first winding. The coiled specimen shall be removed from the cold chamber after a period of 4 hours and brought to room temperature. No readily visible derangements shall result from the cooling or bending. When the cable specimen reaches room temperature, the dielectric strength of the coiled cable specimen shall be checked in accordance with 4.4.4.2.1 to determine its compliance with the requirements of 3.4.1.

Table V. Cold Bend Test Conditions

AWG Size	32	30	28	26	24	22	20	18	16
Mandrel Diameter (Inches)	5/16	3/8	3/8	7/16	1/2	9/16	5/8	3/4	7/8
Test Load (Pounds)	0.4	0.6	0.8	1.0	1.6	6.0	8.0	10.0	16.0

4.5.3.2 Heat resistance. - A cable specimen 30 inches shall be used for this test, and the ends of the cable specimen shall be cut so that the conductor stranding is flush with the conductor insulation. The cable specimen shall be subjected to a temperature of 245 to 255 degrees centigrade (473 to 491 degrees Fahrenheit) for 96 hours in an air oven. At the end of the 96-hour period, the cable specimen shall be removed from the oven and

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allowed to cool to room temperature. One hour after reaching room temperature, the cable specimen shall be wound 5 close turns around a mandrel having diameter 3 times the diameter of the specimen over the jacket, and

(a) The dielectric strength of the cable specimen thus coiled shall be checked in accordance with 4.4.4.2.1 to determine compliance with the requirements of 3.4.2(a),

(b) Any shrinkage or elongation of conductor insulation with respect to conductor stranding for the cable specimen thus coiled shall be measured to determine compliance with the requirements of 3.4.2.(b),

(c) The color of the conductor insulation and of the jacket shall be checked in accordance with 4.5.1.1.8 and 4.5.1.3.1 to determine compliance with the requirements of 3.4.2(c).

4.5.3.3 Soldering resistance. - Each conductor shall be subjected to the soldering test procedure of Specification MIL-W-16878 to determine compliance with the requirements of 3.4.3.

4.5.3.4 Abrasion resistance. - This test shall be performed on each conductor using an abrasion resistance tester which complies with the requirements of Specification MIL-T-5438, with the following exceptions:

(a) Abradant. - 1 inch by 50 yard abradant tape, 400-grit aluminum oxide,
(b) Fault detector sensitivity. - 1 milliampere minimum,
(c) Conducting strip. - Conducting silver paint marked every inch.

A 36-inch specimen of insulated conductor shall be removed from the cable, and shall be drawn between the thumb and finger of one hand by the other hand to straighten it. One inch of insulation shall then be removed from one end of the conductor specimen, and the conductor specimen shall be clamped in the abrasion resistance tester at this end. Prior to clamping the other end, the tension specified in table VI shall be applied to the conductor specimen to make it taut, and it shall be positioned (with the weight support bracket removed) so that it is tangent to the abradant tape.

Using the weight and weight support bracket type specified in table VI, the abradant tape shall be run until failure of the insulation (establishment of conduction) occurs, and the number of inches of tape travel to failure shall be observed and recorded. Seven additional readings of inches of tape travel to failure shall be taken, advancing

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the specimen 2 inches and rotating it 90 degrees before taking each additional reading. The arithmetic average of the eight readings, and the arithmetic average of the readings which are less than the arithmetic average of all eight readings shall be computed to determine compliance with the requirements of 3.4.4.

4.5.3.5 Shock-transportation, storage, and handling. - The complete pack, prepared as specified in section 5, shall be subjected to the appropriate rough handling tests specified in Specification MIL-W-21927.

4.6 Waiver of tests. - Tests specified in 4.5.3.5 may be waived by the procuring agency for level C packs only, provided the contractor guarantees safe delivery to the using activity.

4.7 Functional tests. - At the conclusion of the tests specified in 4.5.3.5, leakage and other tests, as applicable, shall be conducted on the unit packages to determine freedom from rupture of seal and compliance with all the requirements of section 5.

Table VI. Abrasion Resistance Test Conditions

Test Parameter	Cable Size (AWG)								
	32	30	28	26	24	22	20	18	16
Wire Tension Pounds	0.05	0.10	0.15	0.25	0.40	0.60	1.0	1.5	2.0
Weight Support Bracket Type	A	A	A	A	A	A	A	B	B
Weight to be Added - Pounds	0.25	0.25	0.25	0.50	0.50	0.50	0.50	0.75	0.75

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A.

5.1.1.1 Cleaning, drying and preservative application. - Not applicable.

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5.1.1.2 Unit packaging. - The cable shall be furnished on spools or reels in compliance with the level A packaging requirements of Specification MIL-C-12000, with both ends of the cable accessible for testing. In any "inspection lot", at least 80 percent of the total footage shall consist of lengths not less than 50 feet long, and none of the lengths of the remaining total footage shall be less than 10 feet long.

5.1.1.3 Intermediate package. - Not applicable.

5.1.1.4 Cushioning. - The unit package and pack shall incorporate sufficient cushioning material, bracing, or other adequate shock absorbing devices, to ensure that the cable will meet all the requirements of section 3 after being tested in accordance with 4.5.3.5.

At the time of packaging or packing, the moisture content of the cushioning material shall be no more than 5 percent. The pH range of the material shall be between 6.0 and 8.0.

5.1.2 Level B. - Not applicable.

5.1.3 Level C.

5.1.3.1 Cleaning, drying and preservative application. - Not applicable.

5.1.3.2 Unit packaging. - The cable shall be furnished on spools or reels in accordance with good commercial practice, with both ends of the cable accessible for testing, and sealed so as to be moisture proof.

5.1.3.3 Intermediate package. - Not applicable.

5.2 Packing.

5.2.1 Number of packages per pack. - The number of unit packages per pack shall be specified in the contract or purchase order.

5.2.2 Level A.

5.2.2.1 Exterior containers. - Cables packaged as described in 5.1.1.2 shall be packed in compliance with the level A packing requirements of Specification MIL-C-12000. The type, grade, style, and size of the box shall be designated by the procuring activity.

5.2.3 Level B.

5.2.3.1 Exterior containers. - Cables packaged as described in 5.1.1.2 shall be packed in compliance with the level B packing requirements of Specification MIL-C-12000. The type, grade, style, and size of the box shall be designated by the procuring activity.

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5.2.4 Level C. - Cables packaged in accordance with 5.1.3.2 shall be packed to afford protection against damage during shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Consolidated Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Special markings. - None, unless otherwise specified.

5.3.2 Normal markings. - In addition to the normal markings required by contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of Specification MIL-C-12000, including serial numbers or manufacturing lot numbers, and shall give the following:

(a) Name of manufacturer
(b) Date of manufacture
(c) Number and date of this specification
(d) Number of conductors
(e) AWG size
(f) Actual length of cable

6. NOTES

6.1 Intended use. - The cable is intended for use in the Angle Solvers Mark 18 Mod 2 and Mod 3, the Fire Control System Mark 112 Mod 1, the Sonar Receiving Sets AN/BQG-1, -2, -3, -4, or other suitable application.

6.2 Ordering data. - Procurement documents should specify the specific title, number, and date of this specification and exceptions to this specification; applicable drawings and other documents; selections of applicable levels of packaging and packing; number of unit containers per pack; guarantee clause (see 6.4); and total footage of each AWG size.

6.2.1 The attention of the contracting officer is invited to the Quality Assurance Provisions and options in Specification MIL-C-12000 and to 4.6 and 5.2.1 of this specification.

6.3 Criteria for use of proper level of preservation, packaging, and packing shall be as follows:

For level A. This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

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For level B. This level shall be used only when it is definitely known that the item will be held in covered storage in overseas locations for 6 months or less or in domestic locations for an indefinite period.

For level C. This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity.

6.4 Guarantee. - The contract or order shall include a requirement for a guarantee stating that any cable which fails to meet the requirements of this specification within one year after delivery because of a defect in material or workmanship or because of the vendor's failure to comply with the requirements of the applicable drawings and specifications shall be replaced by the vendor at no cost to the Government.

6.5 The contract or order shall include a sufficient number of extra items to cover production and pre-production testing samples.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented inventions that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET
NAVWEPS FORM 4121/3 (8-62)

FORM APPROVED
BUREAU NO. 48-R300

INSTRUCTIONS

This sheet is to be filled out by Government or contractor personnel involved in the use of this specification in procurement of products for ultimate use by the activity shown on the reverse of this sheet.

This sheet is provided for obtaining information which will aid the activity shown in insuring that through use of this specification suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this sheet will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to the activity shown.

SPECIFICATION *(No. and abbrev. title)*

ORGANIZATION	CITY	STATE

CONTRACT NUMBER	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THIS SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
(If so, give paragraph number and wording, and recommendations for correcting the deficiencies.)

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO IF YES, IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If additional papers are attached, send this form and papers in an envelope. This form is addressed for use in window envelope when appropriate.)*

SUBMITTED BY *(Print name and title)*	DATE

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